

The effect of alkalinity enrichment on coral larval settlement

September 2025 pilot trial

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Table of contents

1	Introduction	1
2	Objectives	3
3	Data sources	3
4	Data processing	4
5	Main findings	4
6	References	4

1 Introduction

The success and continued feasibility of coral restoration efforts necessitates they scale to meet the magnitude of the ecological crisis. Addressing this challenge requires approaches that accelerate the production of coral outplants while reducing cost and effort. Although hobby aquarists commonly enrich seawater alkalinity to stimulate coral growth, this practice has rarely been examined in the scientific literature and many questions remain regarding effectiveness on species targeted for restoration and at early life stages.

Restoration practitioners are increasingly prioritizing the sexual production of corals, which enables the deliberate crossing of desirable genotypes, enhances genetic diversity, and supports the creation of more resilient outplants. This research examines whether alkalinity enrichment

can be integrated into this process, with a particular focus on its effects on larval settlement and early post-settlement development.

Gametes from two Caribbean coral species, *Orbicella faveolata* and *Pseudodiploria strigosa*, both commonly targeted for restoration, were collected offshore of Key Largo on 15 August 2025 and transported to the Rosenstiel School of Marine and Atmospheric Science on Virginia Key for fertilization (Figure 1). Once larvae reached competency, they were transferred into 8-oz jars assigned to four treatments (ambient control and three elevated-alkalinity levels). Larvae were allowed to settle on ceramic tiles, after which settlement was quantified.

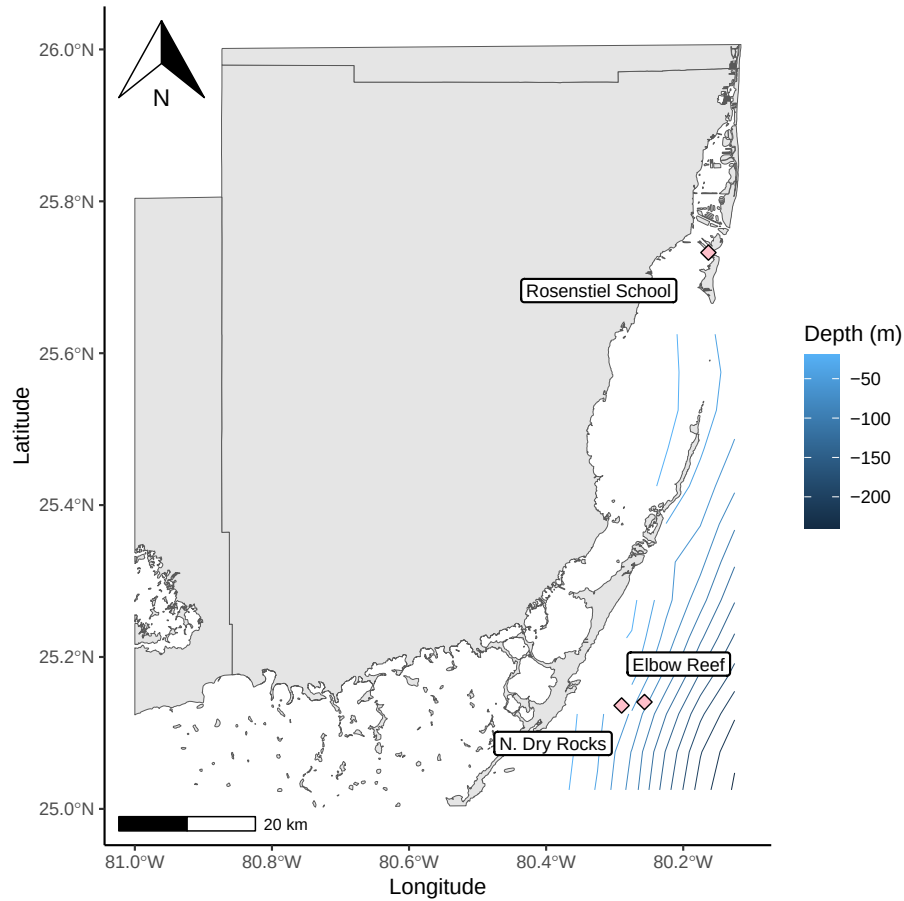


Figure 1: Coral gametes were collected from reefs off of Key Largo, then transported and fertilized at the Rosenstiel School.

2 Objectives

1. Evaluate the feasibility and stability of water-chemistry manipulation for alkalinity enrichment.
2. Assess the ability to induce and quantify larval settlement under controlled laboratory conditions.
3. Determine settlement success across multiple alkalinity treatments.
4. Identify and resolve practical challenges in the experimental setup, including dosing, monitoring, and replication logistics.

3 Data sources

Alkalinity water-chemistry data

Water samples were analyzed using a Metrohm robotic alkalinity titrator, and raw data were exported directly from the Tiamo software. Datasets and variables include:

1. Sample metadata (generated by KMC):
 - “bottle.csv”
2. Raw output from Tiamo software:
 - “settlement_alk_data.csv”

Coral metadata and settlement counts

1. Settlement data:
 - “settlement_data.csv”

Map figure data

1. Florida county geopackage:
 - <https://gis-fdot.opendata.arcgis.com/datasets/florida-county-boundaries-with-fdot-districts/explore>
2. Reef coordinates:
 - “reefs” tribble

```
# A tibble: 3 x 4
  reef          lat    lon species
  <chr>        <dbl> <dbl> <chr>
1 N. Dry Rocks    25.1 -80.3 Orbicella faveolata
2 Elbow Reef      25.1 -80.3 Pseudodiploria strigosa
3 Rosenstiel School 25.7 -80.2 NA
```

4 Data processing

Water chemistry data The raw output from Tiamo software, “settlement_alk_data.csv”, was first cleaned and processed. The sample ID field was standardized by trimming it to retain only the bottle number, allowing it to be accurately joined to the bottle metadata. Because each sample had two replicate alkalinity measurements, one final TA and standard deviation value were created by averaging the replicates using `mutate()` to generate a new column. The final processed dataset retained only the cleaned bottle ID, final TA, TA standard deviation per bottle sample.

The processed output was then joined to the bottle metadata. Treatment statistics were generated using `group()` and `summarize()`.

Coral settlement data

5 Main findings

6 References